PROMPT ENGINEERING

1. What is the advantage of using {user_input} instead of hard-coding input?

Answer: Using {user_input} makes a prompt dynamic and reusable. Instead of writing a fixed value directly into the prompt, the program can accept different inputs from the user at runtime. This makes the application more flexible, easier to maintain, and useful for many different users and situations.

2. When would .format() be enough?

Answer: .format() is enough when the prompt is fairly simple and only needs a small number of variables inserted into a fixed text template. It is a good choice when the prompt does not require complex logic, loops, conditions, or advanced formatting. For example, a prompt such as “Write a summary about {} for {}” can be easily filled using .format(topic, audience).

3. When might Jinja2 be useful?

Answer: Jinja2 is useful when prompts become more complex and need template features such as conditions, loops, reusable blocks, or many variables. It is especially helpful for applications that generate prompts automatically from structured data. For example, Jinja2 can create different prompt sections depending on the user's selected task, language, or output format.

4. Why should prompts be versioned?

Answer: Prompts should be versioned so that changes can be tracked and compared over time. Versioning makes it possible to identify which prompt produced better results, restore an earlier prompt if a new version performs poorly, and collaborate safely with others. It also improves reproducibility because the exact prompt used for an experiment or application can be recorded.

5. What is a prompt regression?

Answer: A prompt regression occurs when a modified prompt performs worse than an earlier version on one or more important tasks. For example, a new prompt may improve answers for one type of question but cause more incorrect, incomplete, or poorly formatted answers for other test cases. Prompt regression testing helps detect these unwanted performance changes before a prompt is released.

6. Why use the same evaluation set when comparing prompt versions?

Answer: The same evaluation set should be used so that different prompt versions are compared under the same conditions. This makes the comparison fair and helps determine whether a change in performance is actually caused by the prompt rather than by different test questions. Using a consistent evaluation set makes results more reliable, measurable, and reproducible.